



CASE STUDY: WANNACRY RANSOMWARE ATTACK ON RENAULT- NISSAN

Cybersecurity for the Operational Technology
Environment (CyOTE)

31 MARCH 2022



U.S. DEPARTMENT OF
ENERGY

Office of
Cybersecurity, Energy Security,
and Emergency Response

This document was prepared by Idaho National Laboratory (INL) under an agreement with and funded by the U.S. Department of Energy.

INL is a U.S. Department of Energy National Laboratory operated by Battelle Energy Alliance, LLC

TABLE OF CONTENTS

- 1. EXECUTIVE SUMMARY 2
- 2. INTRODUCTION 3
 - 2.1. APPLYING THE CYOTE METHODOLOGY 3
 - 2.2. BACKGROUND ON THE ATTACK 5
- 3. OBSERVABLE AND TECHNIQUE ANALYSIS 8
 - 3.1. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR INITIAL ACCESS 8
 - 3.2. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL 9
 - 3.3. EXECUTION THROUGH API TECHNIQUE (T0871) FOR EXECUTION 10
 - 3.4. NETWORK CONNECTION ENUMERATION TECHNIQUE (T0840) FOR DISCOVERY 11
 - 3.5. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR LATERAL MOVEMENT 12
 - 3.6. REMOTE SERVICES TECHNIQUE (T0886) FOR LATERAL MOVEMENT 13
 - 3.7. LATERAL TOOL TRANSFER TECHNIQUE (T0867) FOR LATERAL MOVEMENT 14
 - 3.8. EXECUTION THROUGH API TECHNIQUE (T0871) FOR EXECUTION 15
 - 3.9. MODIFY PROGRAM TECHNIQUE (T0889) FOR PERSISTENCE 16
 - 3.10. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION 17
 - 3.11. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT 18
 - 3.12. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT 19
 - 3.13. CONNECTION PROXY TECHNIQUE (T0884) FOR COMMAND AND CONTROL 20
 - 3.14. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION 21
- APPENDIX A: OBSERVABLES LIBRARY 23
- APPENDIX B: ARTIFACTS LIBRARY 27
- REFERENCES 35

FIGURES

- FIGURE 1. CYOTE METHODOLOGY 3
- FIGURE 2. WANNACRY CYBER-ATTACK TIMELINE 5
- FIGURE 3. WANNACRY ATTACK GRAPH 22

TABLES

- TABLE 1. TECHNIQUES USED IN THE WANNACRY CYBER-ATTACK 7
- TABLE 2. CASE STUDY QUANTITATIVE SUMMARY 7

CASE STUDY: WANNACRY RANSOMWARE ATTACK ON RENAULT-NISSAN

1. EXECUTIVE SUMMARY

The WannaCry Ransomware Attack on Renault-Nissan Case Study leverages publicly available information about the 2017 WannaCry malware campaign and catalogues anomalous observables for each technique employed in the attack. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment ([CyOTE](#) [program](#)).

On 12 May 2017, Renault-Nissan, an international car manufacturing company, was the victim of a WannaCry ransomware attack that resulted in the shutdown of five manufacturing plants. WannaCry displayed ransom notes on infected machines, demanding a payment of \$300 in Bitcoin, which would double to \$600 in three days, to recover encrypted files per machine. The company decided to halt operations and to isolate communications for the infected manufacturing facilities, which were in France, Slovenia, Romania, and India. The losses Renault-Nissan suffered, both from any ransom payments and lost revenues, were not made public. For purposes of comparison, Taiwan Semiconductor Manufacturing Company (TSMC) lost three percent of its quarterly revenues from a WannaCry attack in 2018 that resulted in a disruption of similar duration. In Renault-Nissan's case, the company restarted its manufacturing operations on 15 May, three days after the ransomware was triggered.¹

Renault-Nissan, however, was not the only victim of the May 2017 WannaCry campaign. The ransomware had widespread impact, compromising 200,000 systems throughout 100,000 organizations in over 150 countries worldwide.² While Symantec estimated that WannaCry inflicted \$4 billion in financial losses worldwide,³ the CyOTE program assesses the true cost of this campaign was likely much greater after the cost of infrastructure changes and actions taken to return to normal operations are taken into account.

Using the MITRE ATT&CK® for Industrial Control Systems framework, researchers and analysts identified 14 techniques employed during the attack. The CyOTE program assesses observables accompanying techniques used prior to the triggering event to identify opportunities to detect malicious activity. If observables accompanying the attack techniques are perceived and investigated prior to the triggering event, earlier comprehension of malicious activity can take place. Eight of the identified techniques used during this WannaCry malware campaign were precursors to the triggering event and were associated with 78 observables. Two of these observables were assessed to have an increased likelihood of being perceived in the two months preceding execution of the ransomware.

The information gathered in this case study contributes to a library of observables tied to a repository of artifacts, data sources, technique detection capabilities, and procedural recipes to support the comprehension of indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector asset owners and operators (AOOs) to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber-attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in Figure 1. CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK[®] Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist AOOs in prioritizing their OT environment visibility investments.

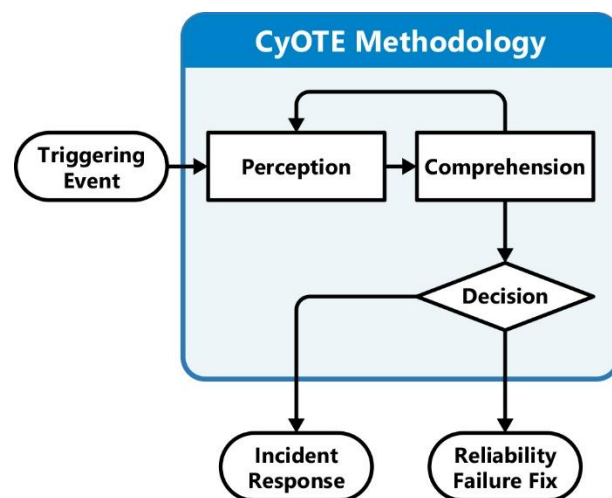


Figure 1. CyOTE Methodology

Case studies such as this one support continued learning through analysis of historical incidents that have impacted OT. This case study is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables AOOs to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the case study's cyber-attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the AOO. The point on this timeline when each technique appears is critical to the AOO's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber-attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for AOOs to detect those observables. If a technique includes effects which AOOs may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides AOOs with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, technique detection capabilities, and procedural recipes to support the comprehension of indicators of attack.

2.2. BACKGROUND ON THE ATTACK

On 12 May 2017, adversaries triggered the WannaCry malware that reportedly resulted in Renault-Nissan shutting down five of their manufacturing plants in India, Romania, Slovenia, and France. While not confirmed in publicly available information, the triggering event was likely the use of the Service Stop technique that could have impacted devices within the OT network if the ransomware had spread from the IT network. Operators would have observed a ransom note on the displays of infected systems, demanding a ransom of \$300 in Bitcoin; whether Renault-Nissan paid the ransom was not publicly disclosed. The affected plants remained offline until 15 May, when they resumed operations.

A timeline of adversarial techniques is shown in Figure 2. The timeline includes the estimated number of days prior to and after the triggering event. The timeline after the triggering event includes the assessed victim comprehension timeline.

On 14 March 2017 (D-60), Microsoft identified and provided a patch for the MS17-010 vulnerability that enables the EternalBlue exploit that was used in the WannaCry attack. The initial release of information regarding the EternalBlue exploit occurred on 14 April.⁴ As a result, the initial access vector for WannaCry is assessed to have occurred on that day, 26 days (D-26) before the Service Stop triggering event on 12 May (D-0).^{5,6}

The first detailed report on the WannaCry malware campaign was released by Poland's Computer Emergency Response Team (CERT) on 15 May (D+3) and is assessed as the comprehension date for the campaign.^{7,8,9,10}

The WannaCry malware campaign included a ransomware component that leverages operating system level exploits in Windows, allowing it to affect enterprise assets. WannaCry had limited criminal success due to the discovery of a killswitch domain by security researchers. The researchers redirected the propagation of the malware within hours of the triggering event to a sinkhole and dampened the impact of the cyber-attack.

The WannaCry malware also created and executed new services, opened backdoors to The Onion Router (TOR) servers, and encrypted dynamic link library files (DLLs). The malware inhibited response efforts by quickly encrypting vulnerable service data and spreading to other vulnerable devices within the network. These actions directly impacted Renault-

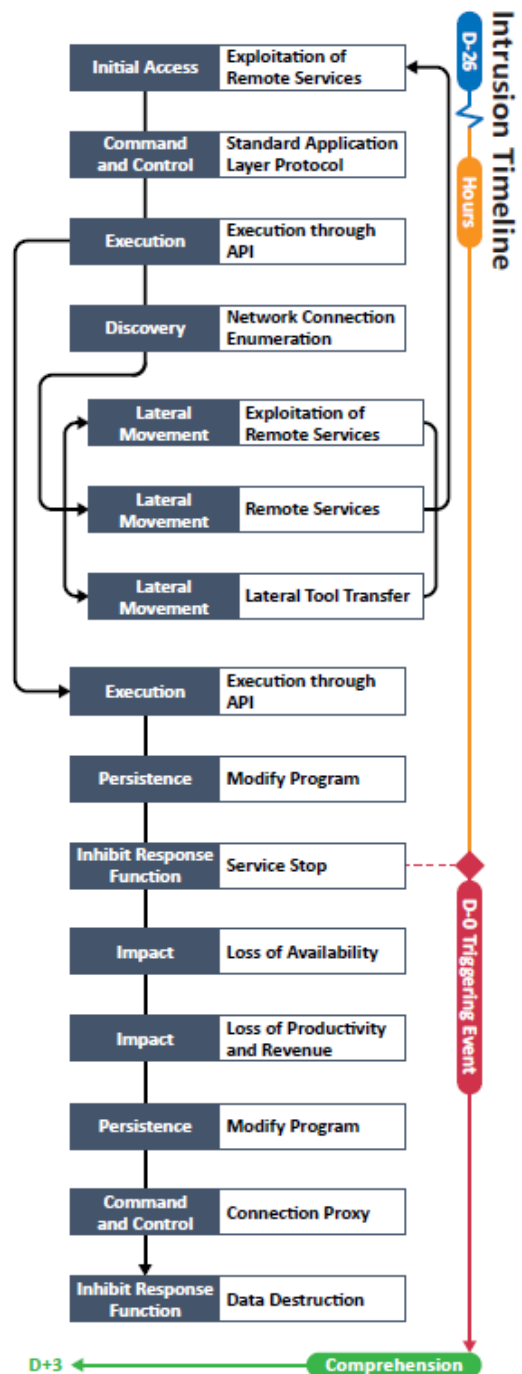


Figure 2. WannaCry Cyber-Attack Timeline

Nissan's ability to function and operate by rendering victim system data inaccessible for a total of three days.^{11,12,13,14}

While details of the WannaCry attack on Renault-Nissan were not made public, analysis by the CyOTE team of how the ransomware functions identified 14 techniques in a sequence and timeframe that were likely used by adversaries during this campaign (Figure 3). These attack techniques are defined according to MITRE's ATT&CK® for Industrial Control Systems framework.

Table 1. Techniques Used in the WannaCry Cyber-Attack

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Data Historian Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Drive-by Compromise	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Engineering Workstation Compromise	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Detect Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
Exploit Public-Facing Application	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Exploitation of Remote Services	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
External Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Internet Accessible Device	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Remote Services	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Replication Through Removable Media	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Rogue Master							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Spearphishing Attachment									Rootkit		Manipulation of View
Supply Chain Compromise									Service Stop		Theft of Operational Information
Wireless Compromise									System Firmware		

Table 2. Case Study Quantitative Summary

Case Study Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	14
Technique Observables	87
Precursor Techniques	10
Precursor Technique Observables	78
Highly Perceivable Precursor Technique Observable	53

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist AOOs in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

3.1. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR INITIAL ACCESS

Adversaries may leverage vulnerabilities within remote services to move between assets or network segments. The adversaries employed the EternalBlue exploit in the Exploitation of Remote Services technique to gain initial access to Renault-Nissan systems on 12 May 2017. Renault-Nissan personnel would have witnessed observables associated with the Exploitation of Remote Services technique within the enterprise and OT environments throughout the entire intrusion.

As part of this technique, WannaCry leverages a vulnerability within the Server Message Block version 1.0 (SMBv1) protocol documented in Microsoft Security bulletin MS 17-010 to gain access to a host. Observers likely would have witnessed initiation of SMB requests via Port 445 on hardcoded IP addresses (192[.]168[.]56.20\IPC\$ and 172[.]16[.]99.IPC\$). WannaCry sets unique values on fields using the SMB protocol, including a Multiplex ID of 81, and failed SMB authentication events could be observed in network traffic.¹⁵

The Exploitation of Remote Services technique (T0866) is responsible for six observables in the IT and OT environments. These observables likely would have been visible to the network administrator and enterprise users, and if identified and investigated earlier in the attack would have reduced comprehension time. This technique is important for investigation because it presents noticeable effects, such as generation of system log entries and anomalous network traffic. This technique also occurs earlier in the timeline and responding to it can effectively halt all future events. Terminating the chain of events at this technique would prevent the use of subsequent techniques.

This technique also allows a malicious actor access to systems beyond the host in question, potentially compromising additional hosts. Of the six observables associated with this technique, five are assessed to be highly perceivable (MS 17-010 Vulnerability within Environment; Initiation of SMB Requests via 445 to Hardcoded IPs 192.168.56.20\IPC\$ and 172.16.99.IPC\$; Unique Values in SMB Fields with Multiplex ID of 81; Failed SMB Authentication; Increase in SMB Authentication Requests).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 31 artifacts could be generated by using the Exploitation of Remote Services technique
Technique Observers^a	OT Staff, OT Cybersecurity, Support Staff, IT Cybersecurity, IT Staff
Resources	Technique Detection Capability Sheet and Recipes

^a Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in Appendix C

3.2. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL

WannaCry also used the EternalBlue exploit to initiate a two-way communication channel via SMB and leveraged the DoublePulsar implant tool to create a backdoor from the victim network to the Internet using TOR. To determine whether to infect a host, the malware would attempt to connect to a killswitch domain. If a connection succeeded, the malware would stop propagating; if the connection failed, the malware would continue to propagate through the victim network.

Observables for this technique have significant overlap with those of the previous technique (T0866). One difference, however, is upon gaining access to a machine, the May 2017 version of WannaCry attempted to connect to this specific killswitch domain, and would continue to propagate until a successful connection could be made:

[http\[.://www\[.\]jiuqerfsodp9ifjaposdfjhgosurijfaewrrergwea.com](http://www[.]jiuqerfsodp9ifjaposdfjhgosurijfaewrrergwea.com)

The Standard Application Layer Protocol technique (T0869) generates five observables in the IT and OT environments. These observables likely would have been visible to the network administrator and enterprise users, and if identified and investigated earlier in the attack would have reduced comprehension time. This technique is important for investigation because it presents noticeable effects, such as generation of system log entries and anomalous network traffic. This technique occurs earlier in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at the Standard Application Layer Protocol technique would limit system damage to anomalous network traffic and system logs.

Of the 5 observables associated with this technique, four are assessed to be highly perceivable (Backdoor Created Using DoublePulsar in Host Memory; Infected Host Connecting to Internet via TOR; Connection Attempts to Malicious URI Across Victim Environment; Outgoing DNS Requests to Killswitch Domain [http\[.://www\[.\]jiuqerfsodp9ifjaposdfjhgosurijfaewrrergwea.com](http://www[.]jiuqerfsodp9ifjaposdfjhgosurijfaewrrergwea.com)).¹⁶

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by using the Standard Application Layer Protocol technique
Technique Observers	OT Staff, OT Cybersecurity, Support Staff, IT Cybersecurity, IT Staff
Resources	Technique Detection Capability Sheet and Recipes

3.3. EXECUTION THROUGH API TECHNIQUE (T0871) FOR EXECUTION

When executed, WannaCry runs as a Windows service (mssecsvc.exe) on the host, executing a function that registers with service handlers to execute as a legitimate service within the Microsoft operating system.

Specific observables IT cybersecurity or network administrators might recognize include the presence of a specific executable invoked binary path name:

BinaryPathName="%currentdirectory%5bef35496fcbdbe841c82f4d1ab8b7c2.exe -msecurity

This registers the mssecsvc.exe executable (MD5 db349b97c37d22f5ea1d1841e3c89eb4) as a service and generates corresponding events (e.g., Event 7036) in the Windows System Log.¹⁷

The Execution Through API technique (T0871) is responsible for at least five observables in the IT and OT environments. These observables likely would have been visible to the network administrator and enterprise users, and if identified and investigated earlier in the attack would have reduced comprehension time. This technique is important for investigation because it presents noticeable effects, such as settings modified within the victim operating system, generation of system log entries, and generation of anomalous network traffic. This technique also occurs earlier in the timeline and responding to it will effectively halt all future events. This technique modifies the victim operating system files, resulting in the host being placed into a modified state. System backups taken after this technique is executed will impact data recovery and disaster recovery efforts.

Of the five observables associated with this technique, four are assessed to be highly perceivable (Malicious Program Runs as Windows Service (mssecsvc.exe) on Host; Malicious Program Executable BinaryPathName="%currentdirectory%5bef35496fcbdbe841c82f4d1ab8b7c2.exe -msecurity."; Mssecsvc.exe Registered Service as MD5 db349b97c37d22f5ea1d1841e3c89eb4; Windows System Log Event 7036 entries).^{18,19}

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 19 artifacts could be generated by using the Execution Through API technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.4. NETWORK CONNECTION ENUMERATION TECHNIQUE (T0840) FOR DISCOVERY

Once the WannaCry malware executes, it starts a service named mssecsvc2.0 (Microsoft Security Center 2.0 Service). This service attempts to create and scan a list of IP ranges on the local network and tries to connect using UDP Ports 137, 138 and TCP Ports 139, 445.^{20,21} This behavior allows WannaCry to enumerate and subsequently propagate among devices throughout the local network.

The malware identifies the subnet on which the system resides, scans IP ranges, connects via SMB, and scans for new file shares every few seconds. WannaCry also enumerates active RDP sessions running on the infected host via the taskse.exe executable:

MD5 2ca2d550e603d74dedda03156023135b38da3630cb014e3d00b1263358c5f00d

WannaCry will attempt to propagate via that protocol in parallel with those accessible via SMB; the enumeration of network drives will result in artifacts within the SMB protocol, as well. At this stage the malware, by consuming extra host or network-level resources, may produce latencies in operational and enterprise devices that could be observable by users.²²

The Network Connection Enumeration technique (T0840) is responsible for at least 12 observables in the IT and OT environments. These observables likely would have been visible to the network administrator and enterprise users, and if identified and investigated earlier in the attack would have reduced comprehension time. This technique is important for investigation because it presents noticeable effects, such as generation of system log entries, increased resource utilization, and generation of anomalous network traffic. This technique also occurs earlier in the timeline and responding to it will effectively halt all future events. This technique enables future techniques access to systems beyond the host in question, resulting in the potential compromise of additional hosts.

Of the 12 observables associated with this technique, seven are assessed to be highly perceivable (Malicious Program Starts Service mssecsvc2.0; Service Starts Scanning Local Subnet for Lists of IP Ranges Over UDP Ports 137, 138 and TCP Ports 139, 445; Internal Network Connections Over SMB Created Every Few Seconds; Host Latency; Network Latency; Attempts to Connect to Random IPs via 445 SMB and 3389 RDP; Host System Log Entries for Session Creation).²³

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 33 artifacts could be generated by using the Network Connection Enumeration technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Staff, Support Staff, IT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.5. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR LATERAL MOVEMENT

For each host in the scanned network identified by WannaCry as having an open SMB service or active RDP session, the malware will create an additional thread to propagate. In the former case, WannaCry leverages the EternalBlue exploit. In the latter, the malware will execute over the RDP session. Specific observables lateral to this technique are similar to those of the previous technique (T0840), although increased network traffic may also be associated with malware propagation and not just network enumeration.

A total of 10 observables were identified with the use of the Exploitation of Remote Services technique (T0866). Seven are assessed to be highly perceivable (Unique Values in SMB Fields; Failed SMB Connections; Remote Session Failures; Host Latency; Network Latency; Host System Log Entries; Attempts to Connect to Random IPs via Port 445 SMB and 3389 RDP).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 31 artifacts could be generated by using the Exploitation of Remote Services technique
Technique Observers	Engineering, OT Staff, IT Staff, Support Staff, OT Cybersecurity, IT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.6. REMOTE SERVICES TECHNIQUE (T0886) FOR LATERAL MOVEMENT

The WannaCry malware will spawn a second thread to propagate to hosts running on networks other than that of the currently infected host. The second thread generates random IPs and attempts to connect to them on Port 445. If the connection is successful, the malware then attempts to exploit the SMBv1 vulnerability mentioned previously. Existing RDP sessions on devices located within external networks are a secondary means through which the malware can propagate outside of its current host's subnet.

Different artifacts would be observable to IT and cybersecurity professionals, depending upon whether the lateral movement was achieved via SMB or RDP. In either case, increases in network traffic for each of these protocols would be visible as the malware simultaneously attempts to connect to multiple external hosts. In the latter case, creating an active RDP session would result in the generation of an Event 4624 in the Windows Event Log along with an unencrypted RDP cookie containing the username. Event 7036, corresponding to changes to services, would also likely appear in the Windows System Log.

A total of eight observables were identified with the use of the Remote Services Technique (T0886). Five are assessed to be highly perceivable (Second Thread Generates Random IPs; Random IPs Attempt to Connect on Port 445 Over SMB; Generation of Windows Event ID: 4624; Unencrypted Cookie Attached to Authentication; Windows System Log Event ID: 7036).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by using the Exploitation of Remote Services technique
Technique Observers	Support Staff, IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.7. LATERAL TOOL TRANSFER TECHNIQUE (T0867) FOR LATERAL MOVEMENT

According to CyOTE analysts, WannaCry leverages both SMB and RDP to move laterally through affected networks. SMB and RDP are necessary components within industrial networks and enable the malware to copy itself over to uninfected hosts. Several of these observables overlap with the previous two techniques, including increased network traffic over the SMB and RDP protocols, the transmission of login credentials, and potential artifacts of creating active sessions (e.g., Event 4624 in the Windows Event Log).

A total of 11 observables were identified with the use of the Lateral Tool Transfer technique (T0867). The observers would likely consist of personnel within Engineering, OT Staff, OT Cybersecurity, Support Staff, IT Cybersecurity, and IT Staff. Seven are assessed to be highly perceivable (Unique Values in SMB Fields; Failed SMB Connections; Remote Session Failures; Host Latency; Network Latency; Host System Log Entries; Attempts to Connect to Random IPs via Port 445 SMB and 3389 RDP).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 21 artifacts could be generated by using the Lateral Tool Transfer technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, Support Staff, IT Cybersecurity, and IT Staff
Resources	Technique Detection Capability Sheet, Recipe

3.8. EXECUTION THROUGH API TECHNIQUE (T0871) FOR EXECUTION

According to CyOTE analysts, after execution, the WannaCry malware carries out an encryption component within the device itself by installing and extracting a PE32 binary (decryptor) from its resource section named “R”.

Once executed, the malware creates and runs `tasksche.exe`, initiating the following process with the `CreateProcess` API:

BinaryPathName="C: \WINDOWS\tasksche.exe / I"

This generates Windows System Log Event 4688 in the event log and Event 7068 in the service log. If the initiated process is run without the “/I” argument, the WannaCry malware loads a XIA resource and decompresses multiple files. Next, the malware evades and obfuscates files by setting hidden attributes for the malware “%CD%” directory.²⁴

A total of eight observables were identified with the use of the Execution through API technique (T0871). The observers would likely consist of personnel within Engineering, OT Staff, OT Cybersecurity, Support Staff, IT Cybersecurity, and IT Staff. Seven are assessed to be highly perceivable (Malicious Software Execute Encryption Component; Installation of PE32 Binary (decryptor); Initiates Process: *BinaryPathName*="C: \WINDOWS\tasksche.exe / I" with the `CreateProcess` API; Windows System Log Event ID: 4688; Windows Service Log Event ID: 7068; Hidden Attribute Set for Malware Directory; Host System Resource Utilization Increases).²⁵

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 19 artifacts could be generated by using the Execution through API technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, Support Staff, IT Cybersecurity, and IT Staff
Resources	Technique Detection Capability Sheet, Recipe

3.9. MODIFY PROGRAM TECHNIQUE (T0889) FOR PERSISTENCE

The WannaCry malware adds a registry key to “tasksche.exe” which modifies the registry.²⁶ To ensure persistence is maintained, the malware creates two registry run keys:

*HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<Randomvalue1>:
<Full_path>\tasksche.exe*

and

*HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<Randomvalue2>:
<Full_path>\tasksche.exe*

Once created, the registry run keys are executed in the task scheduler.

A total of four observables were identified with the use of the Modify Program technique (T0889). All four of these observables are assessed to be highly perceivable (Host System Registry Modification; Host System Registry Key Added; Two Registry Keys Added: “HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<Randomvalue1>: <Full_path>\tasksche.exe”, “HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<Randomvalue2>: <Full_path>\tasksche.exe”).²⁷

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 3 artifacts could be generated by using the Modify Program technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.10. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION

At this point, the WannaCry malware would stop the MS Exchange and Structured Query Language (SQL) servers, potentially impacting any data server (e.g., data historian). As a result, all services would have been shut down manually by operators due to an inability to control or change operations or functions.²⁸ In addition, the process for the decryptor (WanaDecryptor) would spawn.

A total of three observables were identified with the use of the Service Stop technique (T0881). Observers of this activity would be personnel in Engineering, Management, IT Staff, IT Cybersecurity, Support Staff, OT Cybersecurity and OT Staff . All three are assessed to be highly perceivable (MS Exchange Server Services Killed; SQL Server Services Killed; WanaDecryptor Process Spawn).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 13 artifacts could be generated by using the Service Stop technique
Technique Observers	Engineering, Management, IT Staff, IT Cybersecurity, Support Staff, OT Cybersecurity and OT Staff
Resources	Technique Detection Capability Sheet, Recipe

3.11. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT

At this stage along the attack timeline, the victim will start to lose access to files and functionality within the system (e.g., Human Machine Interface (HMI), historian, Programmable Logic Controllers (PLCs), Remote Terminal Units (RTUs), share drive, etc.), resulting in a loss of host and network availability. This could affect industrial applications resulting in a loss of availability, loss of control, and loss of ability to operate. Operators will likely note an initial increase in alarms while simultaneously experiencing a sudden loss of system availability. As industrial applications become unavailable due to the malware encrypting of file shares and/or drives; remote services become unavailable. While operators and engineers work to re-establish process control, a ransom note is identified within the OT environment, providing confirmation of a cyberattack.

A total of seven observables were identified with the use of the Loss of Availability Technique (T0826). Observers of this activity would be personnel in Engineering, Management, IT Staff, IT Cybersecurity, Support Staff, OT Cybersecurity and OT Staff. All seven of these observables are assessed to be highly perceivable (Loss of Access to Files; Loss of Functionality within System; Loss of Access to Share Drive; Loss of Host in Network Availability; Increase in Alarms; Remote Services Become Unavailable; Ransom Note Appears on Workstations).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 8 artifacts could be generated by using the Loss of Availability technique
Technique Observers	Engineering, Management, Support Staff, IT Staff, IT Cybersecurity, OT Staff, and OT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.12. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT

Immediately following the loss of availability of operational systems, the user has lost all access to files and functionality within the system (e.g., HMI, historian, PLCs, RTUs, share drive, etc.), resulting in a loss of trust and confidence in the control and safety systems. Operators experiencing a loss of operational process control will likely be required to troubleshoot the issue by resetting/restarting processes. Operators would find industrial applications inaccessible due to the encryption of file shares and/or drives and a loss of access to remote services.

A total of two observables were identified with the use of the Loss of Productivity and Revenue technique (T0828). Observers of this activity would be personnel in Engineering, Management, IT Staff, IT Cybersecurity, Support Staff, OT Cybersecurity and OT Staff Both are assessed to be highly perceivable (Operations Halted; Plant Output Diminished).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by using the Loss of Productivity and Revenue technique
Technique Observers	Engineering, Management, Support Staff, IT Staff, IT Cybersecurity, OT Staff, and OT Cybersecurity
Resources	Technique Detection Capability Sheet, Recipe

3.13. CONNECTION PROXY TECHNIQUE (T0884) FOR COMMAND AND CONTROL

With operations effectively stopped, the WannaCry malware would continue to maintain its Command-and-Control connections through a TOR proxy.²⁹ Once the ransom is paid, the attackers would use the TOR proxy to deliver the decryption key using traffic via TCP Port 9050 within the infected environment. The individuals who would see this activity are network administrators and cyber defenders.

A total of three observables were identified with the use of the Connection Proxy Technique (T0884). Observers of this activity would be personnel in IT Staff, IT Cybersecurity, OT Cybersecurity and OT Staff All three are assessed to be highly perceivable (Connections with TOR Server via Port 9050; File Decryption Key Delivered via TCP Port 9050; Increase in CPU Resource Usage).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 6 artifacts could be generated by using the Connection Proxy technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity, OT Staff
Resources	Technique Detection Capability Sheet, Recipe

3.14. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION

In the event the ransom is not paid, the attackers could delete the locations where Windows operating systems store copies of files (i.e., Volume Shadow Copy files). The volume shadow copies used for back-ups would be “present” but “unavailable” due to the lack of data, denying the victim the ability to restore files.

A total of three observables were identified with the use of the Data Destruction Technique (T0809). Observers of this activity would be personnel in Engineering, Management, IT Staff, IT Cybersecurity, Support Staff, OT Cybersecurity and OT Staff. All three are assessed to be highly perceivable (Deletion of Volume Shadow Copies; Failed Restorations Attempts; Missing Files).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 27 artifacts could be generated by using the Data Destruction technique
Technique Observers	Engineering, Management, Support Staff, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff
Resources	Technique Detection Capability Sheet, Recipe

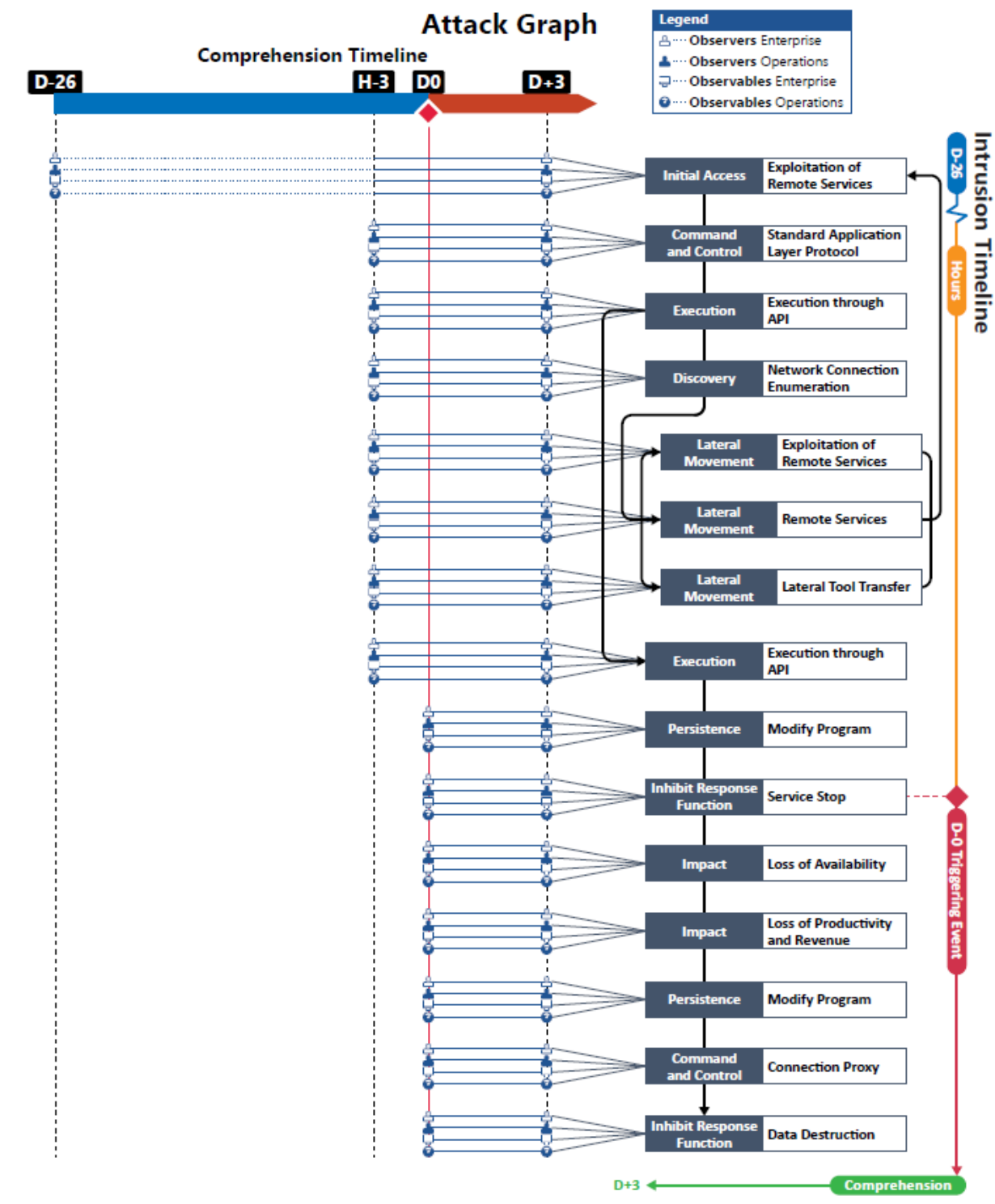


Figure 3. WannaCry Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

Observables Associated with Exploitation of Remote Services Technique (T0866)	
Observable 1	Usage of Remote Services Protocols Between Assets and Network Segments
Observable 2	MS 17-010 Vulnerability Within Environment
Observable 3	Initiation of SMB Requests via 445 to Hardcoded IPs 192.168.56.20\IPC\$ and 172.16.99.IPC\$
Observable 4	Unique Values in SMB Fields with Multiplex ID Of 81
Observable 5	Failed SMB Authentication.
Observable 6	Increase In SMB Authentication Requests

Observables Associated with Standard Application Layer Protocol Technique (T0869)	
Observable 1	SMBv1 Connections
Observable 2	Backdoor Created Using Doublepulsar in Host Memory
Observable 3	Infected Host Connecting to Internet via TOR
Observable 4	Outgoing DNS Requests to Killswitch Domain http[:]//Www[.]luqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com
Observable 5	Connection Attempts to Malicious URI Across Victim Environment

Observables Associated with Execution Through API Technique (T0871)	
Observable 1	Malicious Program Runs as Windows Service (mssecsvc.exe) on Host
Observable 2	Malicious Program Executes Service Handlers as Legitimate Service on Host OS
Observable 3	Malicious Program Executable BinaryPathName="%currentdirectory%5bef35496fcbdbe841c82f4d1ab8b7c2.exe -msecurity."
Observable 4	mssecsvc.exe Registered Service as MD5 db349b97c37d22f5ea1d1841e3c89eb4
Observable 5	Windows System Log Event 7036 Entries

Observables Associated with Network Connection Enumeration Technique (T0840)	
Observable 1	Increase in Threads Running
Observable 2	Malicious Program Starts Service mssecsvc2.0
Observable 3	Service Starts Scanning Local Subnet for Lists of IP Ranges Over UDP Ports 137, 138 and TCP Ports 139, 445
Observable 4	Internal Network Connections Over SMB Created Every Few Seconds
Observable 5	Enumeration of Active RDP Sessions
Observable 6	Scheduled Task Executable taskse.exe Will Attempt to Move to Hosts Sharing RDP Connections

Observables Associated with Network Connection Enumeration Technique (T0840)

Observable 7	Increase in Host Resource Usage
Observable 8	Increase in Network Resources Usage
Observable 9	Host Latency
Observable 10	Network Latency
Observable 11	Attempts to Connect to Random IPs via 445 SMB and 3389 RDP
Observable 12	Host System Log Entries for Session Creation

Observables Associated with Exploitation of Remote Services Technique (T0866)

Observable 1	Exploitation of MS 17-010 Vulnerability Using EternalBlue
Observable 2	Connections on TCP Port 445 SMB
Observable 3	Unique Values in SMB Vields
Observable 4	Failed SMB Connections
Observable 5	Remote Session Connection on TCP 3389 RDP
Observable 6	Remote Session Failures
Observable 7	Host Latency
Observable 8	Network Latency
Observable 9	Host System Log Entries
Observable 10	Attempts to Connect to Random IPs via Port 445 SMB and 3389 RDP

Observables Associated with Remote Services Technique (T0886)

Observable 1	A Second Thread Is Created on Hosts
Observable 2	Second Thread Generates Random IPs
Observable 3	Random IPs Attempt to Connect on Port 445 Over SMB
Observable 4	Existing RDP Sessions Connect to Current Host Subnet
Observable 5	Increase in Network Traffic
Observable 6	Generation of Windows Event ID: 4624
Observable 7	Unencrypted Cookie Attached to Authentication
Observable 8	Windows System Log Event ID: 7036

Observables Associated with Lateral Tool Transfer Technique (T0867)

Observable 1	Use of Login Credential via File Share and Remote Session Services
Observable 2	Host Event ID 4624 Logs for Successful Logons
Observable 3	Connections on TCP Port 445 SMB
Observable 4	Unique Values in SMB Fields

Observables Associated with Lateral Tool Transfer Technique (T0867)

Observable 5	Failed SMB Connections
Observable 6	Remote Session Connection on TCP 3389 RDP
Observable 7	Remote Session Failures
Observable 8	Host Latency
Observable 9	Network Latency
Observable 10	Host System Log Entries
Observable 11	Attempts to Connect to Random IPs via Port 445 SMB and 3389 RDP

Observables Associated with Execution Through API Technique (T0871)

Observable 1	Malicious Software Executes Encryption Component
Observable 2	Installation of PE32 Binary (Decryptor)
Observable 3	Creates and Runs Schedule Task tasksche.Exe
Observable 4	Initiates Process: BinaryPathname="C: \WINDOWS\tasksche.exe / L" with the <i>createprocess</i> API
Observable 5	Windows System Log Event ID: 4688
Observable 6	Windows Service Log Event ID: 7068
Observable 7	Hidden Attribute Set for Malware Directory
Observable 8	Host System Resource Utilization Increases

Observables Associated with Modify Program Technique (T0889)

Observable 1	Host System Registry Modification
Observable 2	Host System Registry Key Added
Observable 3	Registry Key Added: "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<Randomvalue1>: <Full_path>\tasksche.exe"
Observable 4	Registry Key Added "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<Randomvalue2>: <Full_path>\tasksche.exe"

Observables Associated with Service Stop Technique (T0881)

Observable 1	MS Exchange Server Services Killed
Observable 2	SQL Server Services Killed
Observable 3	WanaDecryptor Process Spawn

Observables Associated with Loss of Availability Technique (T0826)

Observable 1	Loss of Access to Files
Observable 2	Loss of Functionality Within System
Observable 3	Loss of Access to Share Drive
Observable 4	Loss of Host in Network Availability
Observable 5	Increase in Alarms
Observable 6	Remote Services Become Unavailable
Observable 7	Ransom Note Appears on Workstations

Observables Associated with Loss of Productivity And Revenue Technique (T0828)

Observable 1	Operations Halted
Observable 2	Plant Output Diminished

Observables Associated with Connection Proxy Technique (T0884)

Observable 1	Connections with TOR Server via Port 9050
Observable 2	File Decryption Key Delivered via TCP Port 9050, if victim paid
Observable 3	Increase in CPU Resource Usage

Observables Associated with Data Destruction Technique (T0809)

Observable 1	Deletion of Volume Shadow Copies
Observable 2	Failed Restorations Attempts
Observable 3	Missing Files

APPENDIX B: ARTIFACTS LIBRARY

Artifacts Associated with Exploitation of Remote Services Technique (T0866)	
Artifact 1	Application Logs
Artifact 2	Connection to HMI End Points
Artifact 3	Connection to EWS End Points
Artifact 4	Connection to Data Historian End Points
Artifact 5	Connection to Controller End Points
Artifact 6	Manipulation of Process
Artifact 7	Manipulation of Set Points
Artifact 8	Misconfigurations of End Points
Artifact 9	Process Failure
Artifact 10	Controller Failure
Artifact 11	Code Injections into Application
Artifact 12	Application Log On Event
Artifact 13	Code Injection into the OS
Artifact 14	OPC Code Injection
Artifact 15	Database Command Executions
Artifact 16	User Events Across Multiple Devices
Artifact 17	Host System Registry Changes
Artifact 18	Security Events Across Multiple Devices
Artifact 19	Kernel Level Events
Artifact 20	System Reboots
Artifact 21	Blank Screens
Artifact 22	Safemode Reboot
Artifact 23	Application Log Off Event
Artifact 24	Alarm Events
Artifact 25	Absence of Alarm Events
Artifact 26	Common Network Traffic
Artifact 27	Remote Network Traffic
Artifact 28	Vendor Specific Network Traffic
Artifact 29	Industrial Protocol Network Traffic
Artifact 30	SQL Protocol
Artifact 31	SMB Protocol

Artifacts Associated with Standard Application Layer Protocol Technique (T0869)	
Artifact 1	External Network Connections
Artifact 2	DNS Autonomous System Number
Artifact 3	Increase in the Number of External Connections
Artifact 4	Network Content Metadata
Artifact 5	Network Connection Times
Artifact 6	HTTP Traffic Port 80
Artifact 7	DNS Traffic Port 53
Artifact 8	SMB Traffic Port 445
Artifact 9	HTTPS Traffic Port 443
Artifact 10	RDP Traffic Port 3389
Artifact 11	HTTP Post Request
Artifact 12	External IP Addresses

Artifacts Associated with Execution Through API Technique (T0871)	
Artifact 1	Process Failure
Artifact 2	Reboot
Artifact 3	Module Load
Artifact 4	Ladder Logic Change
Artifact 5	Timestamps Associated with Activity
Artifact 6	IP Addresses from Network Traffic
Artifact 7	API Log Event (if enabled)
Artifact 8	OPC DA Protocol Network Traffic
Artifact 9	Data Sent with Large File Size
Artifact 10	Data Received with Large File Size
Artifact 11	Network Traffic with Command Execution Content
Artifact 12	State Change in the Process
Artifact 13	Function Execution
Artifact 14	Common Network Traffic
Artifact 15	Industrial Network Traffic
Artifact 16	Vendor Specific Network Traffic
Artifact 17	Remote Connections
Artifact 18	Controller Failure
Artifact 19	Controller Configuration Change

Artifacts Associated with Network Connection Enumeration Technique (T0840)	
Artifact 1	Common Network Traffic
Artifact 2	Polling Network Traffic from Abnormal IP Sender Addresses
Artifact 3	NetBIOS Name Services Port 137
Artifact 4	LDAP Port 389
Artifact 5	Active Directory Calls
Artifact 6	Email Server Calls
Artifact 7	SMTP Port 25 Traffic
Artifact 8	DNS Lookup Queries
Artifact 9	ARP Scans
Artifact 10	TCP Connect Scan
Artifact 11	TCP SYN Scans
Artifact 12	Industrial Network Traffic
Artifact 13	TCP FIN Scans
Artifact 14	TCP Reverse Ident Scan
Artifact 15	TCP XMAS Scan
Artifact 16	TCP ACK Scan
Artifact 17	VNC Port 5900 Calls
Artifact 18	Protocol Content Enumeration
Artifact 19	Protocol Header Enumeration
Artifact 20	Recurring Protocol SYN Traffic
Artifact 21	Sequential Protocol SYN Traffic
Artifact 22	Statistical Anomalies in Network Traffic
Artifact 23	Echo Port 8 Traffic
Artifact 24	Device Failure
Artifact 25	Device Reboots
Artifact 26	Bandwidth Degradation
Artifact 27	Host Recent Connection Logs
Artifact 28	ICMP Port 7 Traffic
Artifact 29	SNMP Port 162 Traffic
Artifact 30	SNMP Port 161 Traffic
Artifact 31	Command Line Dialog Box Open
Artifact 32	Operating System Queries
Artifact 33	DNS Port 53 Zone Transfers

Artifacts Associated with Modify Program Technique (T0889)

Artifact 1	Unexpected Program Download Observed on Network
Artifact 2	Modification to Application Responsible for Program Downloads
Artifact 3	Unexpected Modification to Program Organizational Units on a Device

Artifacts Associated with Remote Services Technique (T0886)

Artifact 1	Remote client connection
Artifact 2	Logoff
Artifact 3	Logoff event
Artifact 4	Registry changes
Artifact 5	Registry connection change
Artifact 6	Mouse movement
Artifact 7	Unexpected I/O
Artifact 8	Desktop prompt windows created
Artifact 9	Session cache
Artifact 10	Application log
Artifact 11	RDP traffic 3389
Artifact 12	Authentication logs
Artifact 13	GUI modifications
Artifact 14	Data file size in network content
Artifact 15	File movement
Artifact 16	MSSQL Traffic 1422 port
Artifact 17	SSH traffic 22
Artifact 18	SMB traffic 139, 445
Artifact 19	VNC traffic 5800, 5900
Artifact 20	Process creation
Artifact 21	Remote session creation timestamp
Artifact 22	Network traffic content creation
Artifact 23	Logon

Artifacts Associated with Loss of Availability Technique (T0826)

Artifact 1	Operator or User Discovery of Encrypted or Inoperable Systems
Artifact 2	Significant Logged Usage of Native Crypto Functions or Presence of Import of Crypto Functions in Binaries
Artifact 3	Significant Reduction or Increase in Network Traffic Due to Malware Propagation of Disappearance of Services

Artifacts Associated with Loss of Availability Technique (T0826)

Artifact 4	Unexplained Loss of Application Data
Artifact 5	Unexplained Loss of User Data
Artifact 6	Process Failure Due to Loss of Required Network or System Dependency
Artifact 7	Changes in Network Routing or Usage of Redundant Control System Network Connection Due to Failed Network Path
Artifact 8	File System Modification Artifacts Might be Associated with the Loss of Availability Might be Present on Disk

Artifacts Associated with Loss of Productivity and Revenue Technique (T0828)

Artifact 1	Loss of Confidence in a Safety System Due to Unreliability Might Result in a Risk Management Driven Shutdown of a Plant
Artifact 2	Wormable or Other Highly Propagating Malware Might Result in the Shutdown of a Plant to Prevent Ransomware or Other Destructive Attacks
Artifact 3	Extortion Attempts Might Lead to Reduced Operations Due to Potential Presence of Malicious Attackers
Artifact 4	Loss of Control of Critical Systems Due to Ransomware or Loss of Confidence Might Lead to a Degraded Productivity or Revenue Operating State
Artifact 5	File System Modification Artifacts Might be Associated with the Loss of Productivity and Revenue Attack Might be Present on Disk

Artifacts Associated with Service Stop Technique (T0881)

Artifact 1	Process Failure
Artifact 2	Alarm Event
Artifact 3	Sysinternals Logs
Artifact 4	Application Error Messages
Artifact 5	Process Error Messages
Artifact 6	Application Service Stop
Artifact 7	OS Service Stop
Artifact 8	System Event Logs
Artifact 9	Application Event Logs
Artifact 10	OS API Call
Artifact 11	Command Line System Argument
Artifact 12	System Resource Usage Manager Application Usage Change
Artifact 13	Registry Change HKLM\System\CurrentControlSet\Services

Artifacts Associated with Connection Proxy Technique (T0884)

Artifact 1	Unexpected Application Communication to Network Proxy Port in Command Line Output (Netstat)
Artifact 2	Unexpected Process Usage of Network Proxy Port Observed via Memory
Artifact 3	Unexpected Process Usage of Network Proxy Port Observed via OS Logs
Artifact 4	Unexpected Process Usage of Network Proxy Port Observed via Firewall Logs
Artifact 5	Unexpected Host Communicating with Network Proxy Port on Industrial Asset
Artifact 6	Unusual Network or Host Communications Identified in Network Proxy Log

Artifacts Associated with Data Destruction Technique (T0809)

Artifact 1	Program Execution
Artifact 2	Telnet Port 23
Artifact 3	SFTP Port 22
Artifact 4	FTPS Port 990
Artifact 5	SMB Port 139, 445
Artifact 6	HTTP Port 80
Artifact 7	HTTPS Port 443
Artifact 8	Command Line Arguments
Artifact 9	SCP Port 22
Artifact 10	Memory Corruption
Artifact 11	Files Moved to Recycle Bin
Artifact 12	Non-Native Files
Artifact 13	Transient Device Connections
Artifact 14	External Network Connections
Artifact 15	Local Network Connections
Artifact 16	Host System Reboot Failure
Artifact 17	Process Logic Failure
Artifact 18	Event Log Creation
Artifact 19	System Call
Artifact 20	System Application Interruption
Artifact 21	Device Failure
Artifact 22	Recovery Attempt Failure
Artifact 23	File Encryptions
Artifact 24	Missing Files
Artifact 25	Use of File Transfer Protocols
Artifact 26	FTP Port 20, 21

Artifacts Associated with Data Destruction Technique (T0809)

Artifact 27	TFTP Port 60
--------------------	--------------

Artifacts Associated with Lateral Tool Transfer (T0867)

Artifact 1	Command Execution
Artifact 2	File Metadata Changes
Artifact 3	User Information Changes
Artifact 4	Process Creation
Artifact 5	System Resource Usage Management Events
Artifact 6	Data Sent from One Location to Another
Artifact 7	Data Received from One Location to Another
Artifact 8	SQL Commands
Artifact 9	SQL Create Commands
Artifact 10	SQL Insert Commands
Artifact 11	Command Prompt Dialog Box Open
Artifact 12	Command Prompt Dialog Box Open
Artifact 13	.dll Injection into File Directory
Artifact 14	.dll Execution
Artifact 15	Powershell Dialog Box Open
Artifact 16	Common Network Traffic
Artifact 17	Remote Network Traffic
Artifact 18	Industrial Network Traffic
Artifact 19	File Creation
Artifact 20	File Modification
Artifact 21	File Deletion

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	• Security Tester
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

REFERENCES

- ¹ [Industrial Cybersecurity Pulse | Gary Cohen | “Throwback Attack: WannaCry ransomware takes Renault-Nissan plants offline” | <https://www.industrialcybersecuritypulse.com/throwback-attack-wannacry-ransomware-takes-renault-nissan-plants-offline/> | 22 April 2021 | Accessed on 16 March 2022 | The source is publicly available information and does not contain classification markings]
- ² [Nick Ismail | “The Real Damage of a Ransomware Attack is Felt in the Downtime” | <https://www.information-age.com/real-damage-ransomware-attack-felt-downtime-123466541/> | 1 June 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ³ [Broadcom | Charles Cooper | “WannaCry: Lessons Learned 1 Year Later” | <https://symantec-enterprise-blogs.security.com/blogs/feature-stories/wannacry-lessons-learned-1-year-later> | 15 May 2018 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ⁴ [Trend Micro | “Shadow Brokers Leaks Hacking Tools: What it Means for Enterprises” | <https://www.trendmicro.com/vinfo/mx/security/news/vulnerabilities-and-exploits/shadow-brokers-leaks-hacking-tools-what-it-means-for-enterprises> | 18 April 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ⁵ [Cybersecurity and Infrastructure Security Agency | “Multiple Ransomware Infections Reported” | <https://www.cisa.gov/uscert/ncas/current-activity/2017/05/12/Multiple-Ransomware-Infections-Reported> | 15 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ⁶ [Poland CERT | Kamil Frankowicz | “WannaCry” | <https://www.cert.pl/posts/2017/05/wannacry-ransomware/> | 15 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ⁷ [Poland CERT | Kamil Frankowicz | “WannaCry” | <https://www.cert.pl/posts/2017/05/wannacry-ransomware/> | 15 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ⁸ [Washington Post | Ellen Nakashima and Craig Timberg | “NSA officials worried about the day its potent hacking tool would get loose, then it did” | https://www.washingtonpost.com/business/technology/nsa-officials-worried-about-the-day-its-potent-hacking-tool-would-get-loose-then-it-did/2017/05/16/50670b16-3978-11e7-a058-ddbb23c75d82_story.html | 16 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ⁹ [The Atlantic | Bruce Schneier | “Who are the Shadow Brokers” | <https://www.theatlantic.com/technology/archive/2017/05/shadow-brokers/527778/> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁰ [Kamil Frankowicz | “WannaCry Ransomware” | <https://frankowicz.me/wannacry-ransomware/> | 15 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹¹ [Cybersecurity and Infrastructure Security Agency | “Alert TA17-132A: Indicators Associated With WannaCry Ransomware” | <https://www.cisa.gov/uscert/ncas/alerts/TA17-132A> | 12 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹² [Industrial Cybersecurity Pulse | Gary Cohen | “Throwback Attack: WannaCry ransomware takes Renault-Nissan plants offline” | <https://www.industrialcybersecuritypulse.com/throwback-attack-wannacry-ransomware-takes-renault-nissan-plants-offline/> | 22 April 2021 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹³ [Reuters | “Renault-Nissan resumes nearly all production after cyber attack” | <https://www.reuters.com/article/us-cyber-attack-renault/renault-nissan-resumes-nearly-all-production-after-cyber-attack-idUSKCN18B0S5> | 15 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]

-
- ¹⁴ [France24 | “France’s Renault hit in worldwide ‘ransomware’ cyber attack” | <https://www.france24.com/en/20170512-cyberattack-ransomware-renault-worldwide-british-hospitals> | 14 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁵ [LogRhythm | Erika Noerenberg, Andrew Costis, and Nathaniel Quist | “A Technical Analysis of WannaCry Ransomware” | <https://logrhythm.com/blog/a-technical-analysis-of-wannacry-ransomware/> | 16 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁶ [LogRhythm | Erika Noerenberg, Andrew Costis, and Nathaniel Quist | “A Technical Analysis of WannaCry Ransomware” | <https://logrhythm.com/blog/a-technical-analysis-of-wannacry-ransomware/> | 16 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁷ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁸ [Vipre | “WannaCry Technical Analysis” | <https://support.threattracksecurity.com/support/solutions/articles/1000250396-wannacry-technical-analysis> | 26 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁹ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²⁰ [LogRhythm | Erika Noerenberg, Andrew Costis, and Nathaniel Quist | “A Technical Analysis of WannaCry Ransomware” | <https://logrhythm.com/blog/a-technical-analysis-of-wannacry-ransomware/> | 16 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²¹ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²² [LogRhythm | Erika Noerenberg, Andrew Costis, and Nathaniel Quist | “A Technical Analysis of WannaCry Ransomware” | <https://logrhythm.com/blog/a-technical-analysis-of-wannacry-ransomware/> | 16 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²³ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²⁴ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²⁵ [Vipre | “WannaCry Technical Analysis” | <https://support.threattracksecurity.com/support/solutions/articles/1000250396-wannacry-technical-analysis> | 26 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²⁶ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²⁷ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]
- ²⁸ [Mandiant | Alex Berry, Josh Homan, and Randi Eitzman | “WannaCry Malware Profile” | <https://www.mandiant.com/resources/wannacry-malware-profile> | 23 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]

²⁹ [Vipre | “WannaCry Technical Analysis” | <https://support.threattracksecurity.com/support/solutions/articles/1000250396-wannacry-technical-analysis> | 26 May 2017 | Accessed on 1 March 2022 | The source is publicly available information and does not contain classification markings]